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The Director

of the United States Patent and Trademark Office has received an application for a patent for a new and useful invention. The title and description of the invention are enclosed. The requirements of law have been complied with, and it has been determined that a patent on the invention shall be granted under the law.

Therefore, this United States

Patent

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John A. Guarnieri

DIRECTOR OF THE UNITED STATES PATENT AND TRADEMARK OFFICE



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If the application for this patent was filed on or after June 8, 1995, the term of this patent begins on the date on which this patent issues and ends twenty years from the filing date of the application or, if the application contains a specific reference to an earlier filed application or applications under 35 U.S.C. 120, 121, 365(c), or 386(c), twenty years from the filing date of the earliest such application (“the twenty-year term”), subject to the payment of maintenance fees as provided by 35 U.S.C. 41(b), and any extension as provided by 35 U.S.C. 154(b) or 156 or any disclaimer under 35 U.S.C. 253.

If this application was filed prior to June 8, 1995, the term of this patent begins on the date on which this patent issues and ends on the later of seventeen years from the date of the grant of this patent or the twenty-year term set forth above for patents resulting from applications filed on or after June 8, 1995, subject to the payment of maintenance fees as provided by 35 U.S.C. 41(b) and any extension as provided by 35 U.S.C. 156 or any disclaimer under 35 U.S.C. 253.



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(54) **DEVICE FOR CENTERING A SENSOR ASSEMBLY IN A BORE**

(71) Applicant: **Petromac IP Limited**, Auckland (NZ)

(72) Inventor: **Stephen Peter McCormick**, Auckland (NZ)

(73) Assignee: **Petromac IP Limited**, Auckland (NZ)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

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E21B 47/01 (2012.01)

(52) **U.S. Cl.**
CPC **E21B 17/1021** (2013.01); **E21B 47/01** (2013.01)

(58) **Field of Classification Search**
CPC E21B 17/1021; E21B 47/01
See application file for complete search history.

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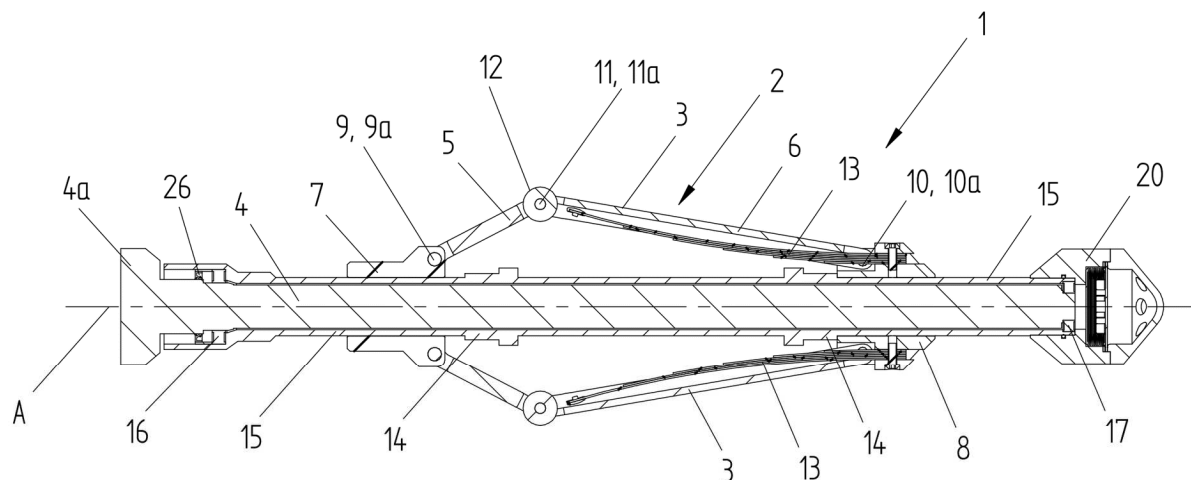
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Primary Examiner — David Carroll
(74) *Attorney, Agent, or Firm* — Niels Haun; Dann, Dorfman, Herrell & Skillman, P.C.

(57) **ABSTRACT**

A centering device is connected to a rotating sensor assembly located at the bottom end of a tool string. The device comprises a centraliser with a plurality of arms spaced circumferentially apart around a longitudinal central axis of the device, and a shaft with a mount for fixing the shaft to the rotating sensor assembly. The shaft extends below the rotating sensor assembly, and the shaft and rotating sensor assembly rotate together on the longitudinal central axis. The centraliser is rotationally mounted to the shaft to allow the shaft to rotate within the centraliser and with the centraliser positioned below the rotating sensor assembly.

20 Claims, 4 Drawing Sheets



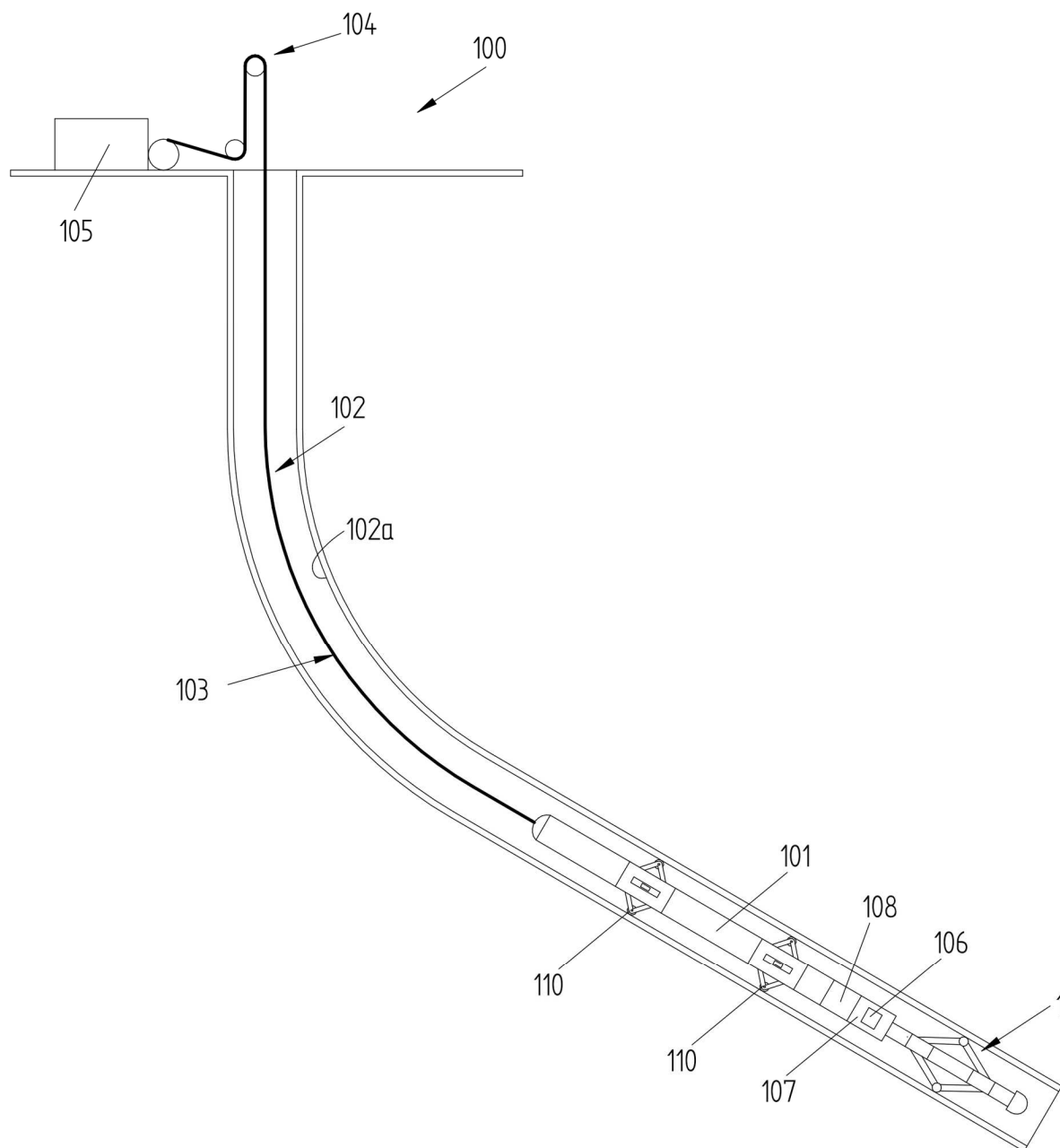


FIGURE 1

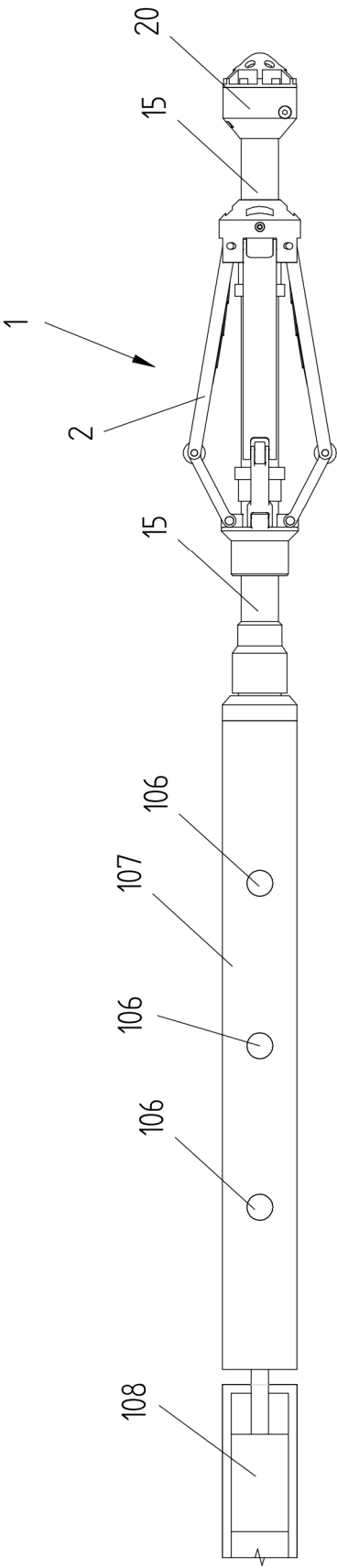


FIGURE 2

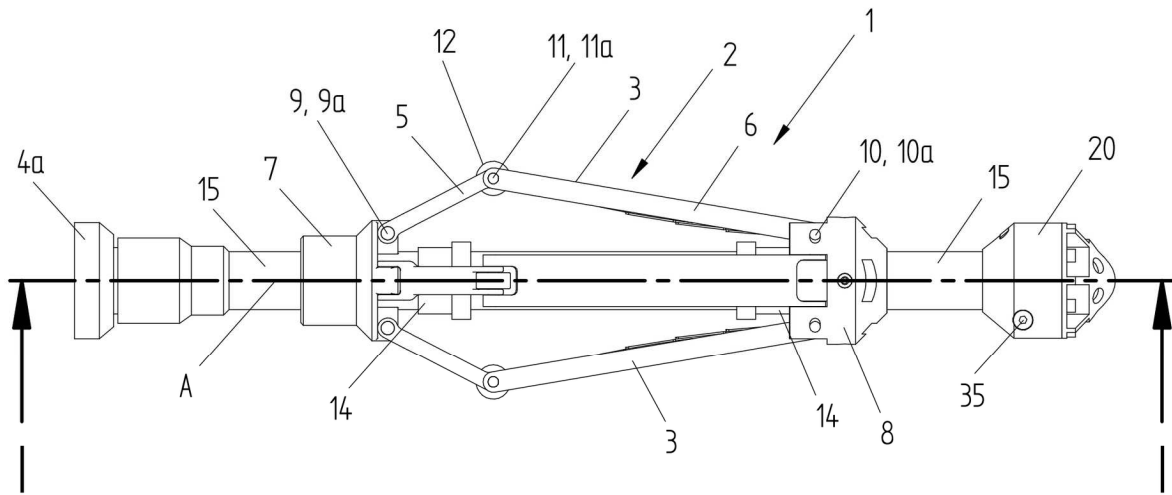


FIGURE 3

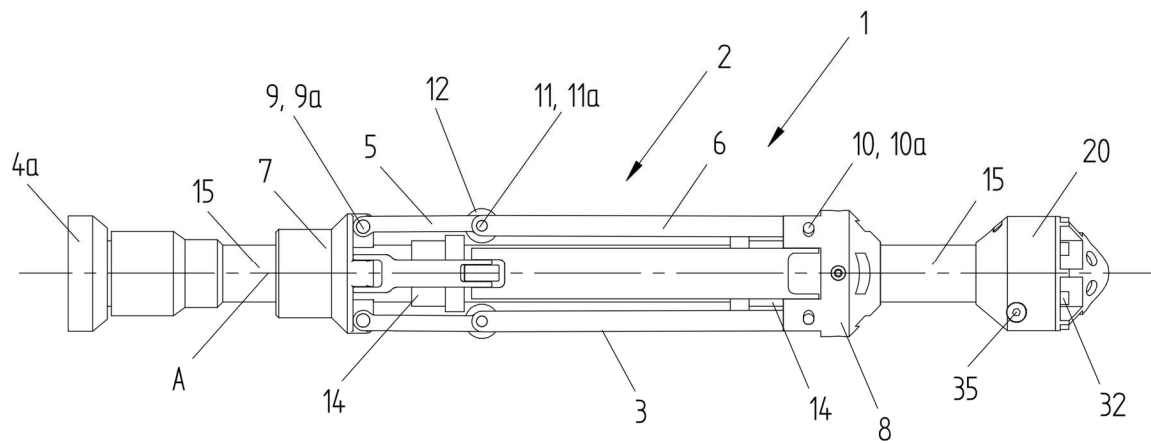


FIGURE 4

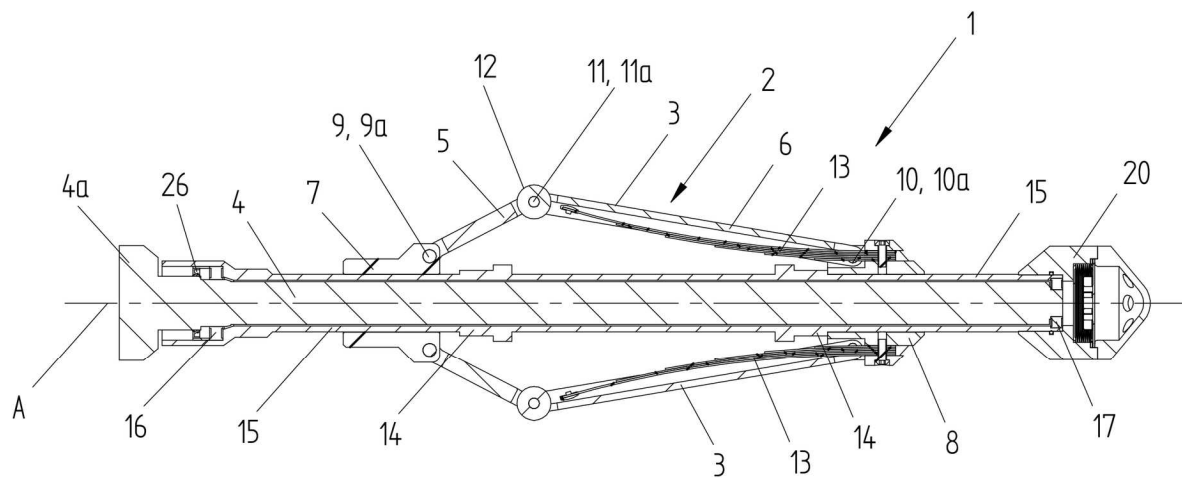


FIGURE 5

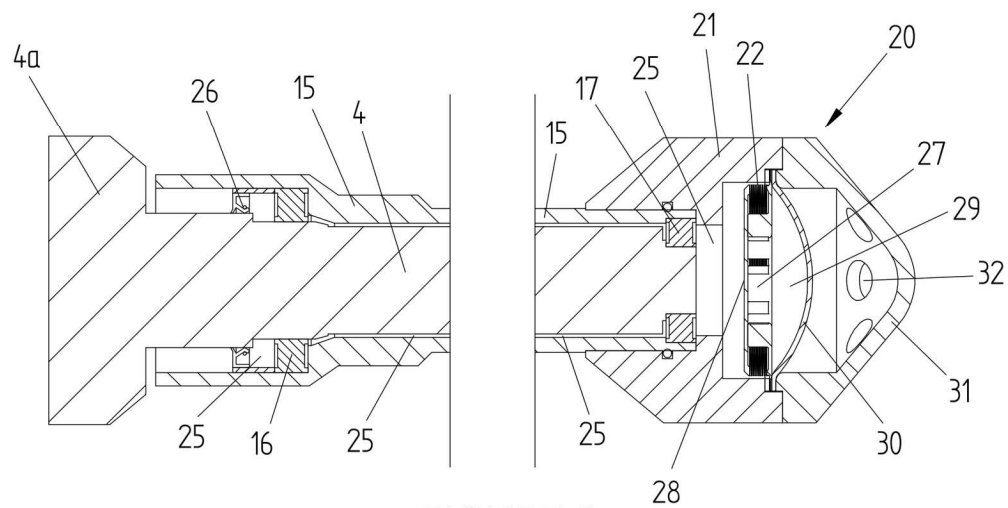


FIGURE 6

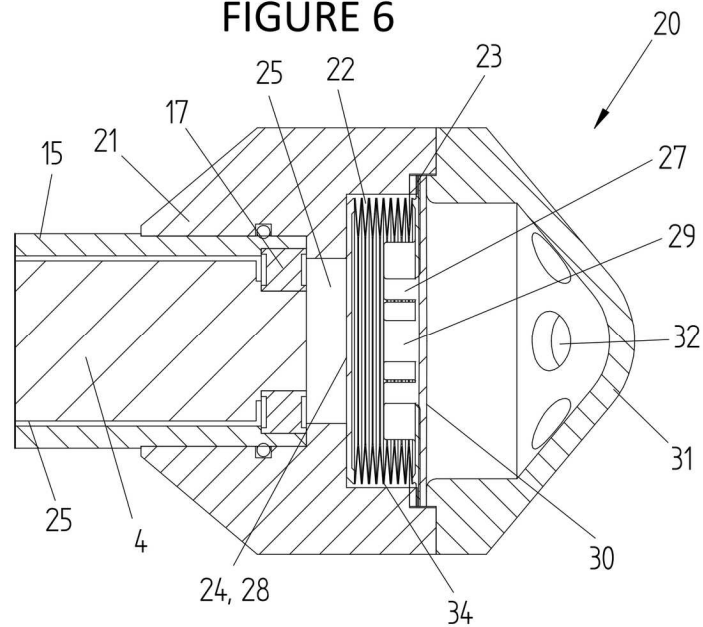


FIGURE 7

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DEVICE FOR CENTERING A SENSOR ASSEMBLY IN A BORE

RELATED APPLICATIONS

This application claims benefit of New Zealand Provisional Patent Application No. 824169, filed on Aug. 11, 2025, entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

This invention relates to devices for use in centering sensor equipment in a bore such as a pipe, a wellbore or a cased wellbore, and in particular to devices for use in centering sensor equipment in wireline logging applications.

BACKGROUND

Hydrocarbon exploration and development activities rely on information derived from sensors which capture data relating to the geological properties of an area under exploration. One approach used to acquire this data is through wireline logging. Wireline logging is performed in a wellbore immediately after a new section of hole has been drilled, referred to as open-hole logging. These wellbores are drilled to a target depth covering a zone of interest, typically between 1000-5000 meters deep. A sensor package, also known as a “logging tool” or “tool-string” is then lowered into the wellbore and descends under gravity to the target depth in the wellbore. The logging tool is lowered on a wireline-being a collection of electrical communication wires which are sheathed in a steel cable connected to the logging tool. The steel cable carries the loads from the tool-string, the cable itself, friction forces acting on the downhole equipment and any overpulls created by sticking or jamming. Once the logging tool reaches the target depth it is then drawn back up through the wellbore at a controlled rate of ascent, with the sensors in the logging tool operating to generate and capture geological data.

Wireline logging is also performed in wellbores that are lined with steel pipe or casing, referred to as cased-hole logging. After a section of wellbore is drilled, casing is lowered into the wellbore and cemented in place. The cement is placed in the annulus between the casing and the wellbore wall to ensure isolation between layers of permeable rock layers intersected by the wellbore at various depths. The cement also prevents the flow of hydrocarbons in the annulus between the casing and the wellbore which is important for well integrity and safety. Oil wells are typically drilled in sequential sections. The wellbore is “spudded” with a large diameter drilling bit to drill the first section. The first section of casing is called the conductor pipe. The conductor pipe is cemented into the new wellbore and secured to a surface well head. A smaller drill bit passes through the conductor pipe and drills the surface hole to a deeper level. A surface casing string is then run in hole to the bottom of the hole. This surface casing, commonly 20" (nominal OD) is then cemented in place by filling the annulus formed between the surface casing and the new hole and conductor casing. Drilling continues for the next interval with a smaller bit size. Similarly, intermediate casing (e.g. 13 $\frac{3}{8}$ ") is cemented into this hole section. Drilling continues for the next interval with a smaller bit size. Production casing (e.g. 9 $\frac{5}{8}$ " OD) is run to TD (total depth) and cemented in place. A final casing string (e.g. 7" OD) is cemented in place from a liner hanger from the previous

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casing string. Therefore, the tool-string must transverse down a cased-hole and may need to pass into a smaller diameter bore.

There is a wide range of logging tools which are designed to measure various physical properties of the rocks and fluids contained within the rocks. The logging tools include transducers and sensors to measure properties such as electrical resistance, gamma-ray density, speed of sound and so forth. The individual logging tools are combinable and are typically connected together to form a logging tool-string. Some sensors are designed to make close contact with the borehole wall during data acquisition whilst others are ideally centered in the wellbore for optimal results. These requirements need to be accommodated with any device that is attached to the tool-string.

In cased hole, logging tools are used to assess the strength of the cement bond between the casing and the wellbore wall and the condition of the casing. There are several types of sensors, and they typically need to be centered in the casing. One such logging tool utilises high frequency ultrasonic acoustic transducers and sensors to record circumferential measurements around the casing. The ultrasonic transmitter and sensor are mounted on a rotating head connected to the bottom of the tool. This rotating head spins and enables the sensor to record azimuthal ultrasonic reflections from the casing wall, cement sheath, and wellbore wall as the tool is slowly winched out of the wellbore. Other tools have transmitters and sensors that record the decrease in amplitude, or attenuation, of an acoustic signal as it travels along the casing wall. It is important that these transducers and sensors are well centered in the casing to ensure that the data recorded is valid. Other logging tools that measure fluid and gas production in flowing wellbores may also require sensor centralisation. Logging tools are also run in producing wells to determine flow characteristics of produced fluids. Many of these sensors also require centralisation for the data to be valid.

In open hole (uncased wellbores), logging tools are used to scan the wellbore wall to determine the formation structural dip, the size and orientation of fractures, the size and distribution of pore spaces in the rock and information about depositional environment. One such tool has multiple sensors on pads that contact the circumference of the wellbore to measure micro-resistivity. Other tools generate acoustic signals which travel along the wellbore wall and are recorded by multiple receivers spaced along the tool and around the azimuth of the tool. As with the cased hole logging tools, the measurement from these sensors is optimised with good centralisation in the wellbore.

A common apparatus to centralise logging tools is a bow-spring centraliser. Bow-spring centralisers incorporate a number of curved leaf springs. The leaf springs are attached at their ends to an attachment structure that is fixed to the logging tool. The midpoint of the curved leaf spring (or bow springs) is arranged to project radially outward from the attachment structure and tool string. When the bow-spring centraliser is not constrained by the wellbore, the outer diameter of the bow-spring centraliser is greater than the diameter of the wellbore or casing in which it is to be deployed. Once deployed in the wellbore, the bow-springs are compressed, and the compressed bow springs provide a centering force on the tool string to hold it centrally in the bore.

Another known type of centraliser consists of several levers or arms with a wheel at or near where the levers are pivotally connected together. There are multiple sets of lever-wheel assemblies disposed at equal azimuths around

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the central axis of the device. There are typically between three and six sets. The ends of each lever set are connected to blocks which are free to slide axially on a central mandrel of the centraliser device. Springs are used force these blocks to slide toward each other forcing the arms to extend radially outward so that the wheels exert force against the wellbore wall. The centraliser device is typically energised by means of either axial or radial acting spring or a combination of both. The advantage of this type of centraliser is that drag is reduced by the wheels which roll, rather than slide along the wellbore wall.

A wireline logging tool-string is typically in the order of 20 ft to 100 ft long and 2" to 5" in diameter. Centralisers may be mounted to and spaced apart along the tool string to carry the tool string centrally in the bore. A centraliser should be placed on the tool string near to any sensor that must be centred in the bore. Even if a centraliser perfectly centres the section of tool string that it is carrying, a sensor in the tool string axially spaced from the centraliser may be off centre in the bore due to flex in the tool string and/or curvature of the wellbore as it transitions from a vertical section to a deviated section.

The reference to any prior art in the specification is not, and should not be taken as, an acknowledgement or any form of suggestion that the prior art forms part of the common general knowledge in any country.

DISCLOSURE OF INVENTION

It is an object of the present invention to address any one or more of the above problems or to at least provide the industry with a useful device for centering sensor equipment in a bore or pipe.

According to one aspect of the present invention there is provided a device for centering a rotating sensor assembly in a wellbore, the rotating sensor assembly located at the bottom end (the terminal end) of a tool string, the device comprising:

- a centraliser comprising a plurality of arms spaced circumferentially apart around a longitudinal central axis of the device,
- a shaft with a mount for fixing the shaft to the rotating sensor assembly to extend below the rotating sensor assembly, so that the shaft and rotating sensor assembly rotate together on the longitudinal central axis,
- wherein the centraliser is rotationally mounted to the shaft to allow the shaft to rotate within the centraliser and with the centraliser positioned below the rotating sensor assembly.

In some embodiments, in use, an upper end of the shaft is connected to the rotating sensor assembly and the lower end of the shaft is a terminal end at the bottom end of the tool string.

In some embodiments, the centraliser supports the shaft between the upper connected end of the shaft and the lower terminal end of the shaft.

In some embodiments, the centraliser comprises a first support member and a second support member, the arms connected between the first and second support members, wherein one or both of the first and second support members are adapted to move axially along the central longitudinal axis, and wherein the first and second support members are rotationally mounted to the shaft.

In some embodiments, the centralising device comprises a sleeve rotationally mounted to the shaft to allow for

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rotation of the shaft within the sleeve, the centraliser mounted to the sleeve to thereby rotationally mount the centraliser to the shaft.

In some embodiments, the device comprises an upper bearing located at or towards an upper end of the shaft and a lower bearing located at or towards the lower end of the shaft, the upper and lower bearings between the shaft and sleeve.

In some embodiments, the device comprises a pressure compensated bearing lubrication system, the pressure compensated bearing lubrication system comprising:

- a housing attached to a lower end of the sleeve,
- a resiliently deformable member attached to the housing, and
- a seal between the rotating shaft and the sleeve at the upper end of the sleeve,
- a sealed volume defined by the resiliently deformable member, an inside of the housing, an OD of the shaft, an ID of the sleeve, and the seal, the upper and lower bearings received in the sealed volume,
- in use the sealed volume filled with lubricant, the resiliently deformable member providing a movable interface between the wellbore fluid and the lubricant received in the sealed volume to pressure compensate the lubricant relative to the ambient pressure of the wellbore.

In some embodiments, the housing with the resiliently deformable member closes an end of the sleeve to cover a lower end of the shaft.

In some embodiments, the pressure compensated bearing lubrication system comprises a cover over the resiliently deformable member, the cover with at least one opening such that well bore pressure is applied to the resiliently deformable member.

In some embodiments, the resiliently deformable member is or comprises a bellows formation.

In some embodiments, the sealed volume is defined by an exterior of the bellows formation, the inside of the housing, the ID of the sleeve, the OD of the rotating shaft, and the seal.

In some embodiments, the pressure compensated bearing lubrication system comprises a flexible member,

- the bellows formation has an open end and a closed end, the flexible member sealingly connected to or over the open end of the bellows formation to form a chamber defined at least in part by an interior of the bellows formation and an inner surface of the flexible member, and

in use the chamber filled with a fluid so that wellbore fluid acting on an external surface of the flexible member is communicated to the bellows formation.

In some embodiments, the centraliser comprises a first support member and a second support member axially spaced apart along a longitudinal axis of the device; and

each arm is an arm assembly, the plurality of arm assemblies connected between the first and second support members, each arm assembly comprising:

- a first arm pivotally connected to the first support member by a first pivot joint,

a second arm pivotally connected to the second support member by a second pivot joint, the first and second arms pivotally connected via a third pivot joint,

wherein one or both of the first and second support members is adapted to move axially along the longitudinal axis to allow the arm assemblies to extend and retract radially with respect to the longitudinal axis.

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In some embodiments, each arm assembly comprises a roller or wheel to contact the wellbore wall. In some embodiments, the roller or wheel rotates around the third pivot joint.

In some embodiments, the centraliser comprises one or more spring elements to bias the arm assemblies radially outwards to contact the bore wall.

In some embodiments, the centraliser is a passive device, with energisation of the arms radially outwards being provided by one or more spring elements of the device only.

Unless the context suggests otherwise, the term “wellbore” may refer to both cased and uncased wellbores. Thus, the term ‘wellbore wall’ may refer to the wall of a wellbore or the wall of a casing within a wellbore.

Unless the context suggests otherwise, the term “tool string” refers to an elongate sensor package or assembly also known in the industry as a “logging tool”, and may include components other than sensors such as guide and orientation devices and carriage devices attached to sensor components or assemblies of the tool string. A tool string may include a single elongate sensor assembly, or two or more sensor assemblies connected together.

Unless the context suggests otherwise, the terms “upper”, “lower”, “top”, “bottom” and similar such terms are used for convenience and ease of explanation. Such terms are not considered to be limiting to any particular orientation of a tool string or device comprising the invention in use. For example, when deployed in a horizontal bore or pipe, the “bottom end” or “lower end” of a tool string or device is the terminal end of the tool string or device. The “top end” is the end of the tool string connected to a conveyance means such as a wireline or drill pipe.

Unless the context clearly requires otherwise, throughout the description and the claims, the words “comprise”, “comprising”, and the like, are to be construed in an inclusive sense as opposed to an exclusive or exhaustive sense, that is to say, in the sense of “including, but not limited to”. Where in the foregoing description, reference has been made to specific components or integers of the invention having known equivalents, then such equivalents are herein incorporated as if individually set forth.

The invention may also be said broadly to consist in the parts, elements and features referred to or indicated in the specification of the application, individually or collectively, in any or all combinations of two or more of said parts, elements or features, and where specific integers are mentioned herein which have known equivalents in the art to which the invention relates, such known equivalents are deemed to be incorporated herein as if individually set forth.

Further aspects of the invention, which should be considered in all its novel aspects, will become apparent from the following description given by way of example of possible embodiments of the invention.

BRIEF DESCRIPTION OF THE DRAWINGS

An example embodiment of the invention is now discussed with reference to the Figures.

FIG. 1 is a schematic representation of a well site and a tool string descending a wellbore in a wireline logging operation.

FIG. 2 is a schematic representation of a centralising device mounted to a rotating sensor assembly at the bottom end of a tool string, according to an embodiment of the present invention.

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FIG. 3 is a side view representation of the centralising device, with arms of a centraliser of the centralising device in a radially outwards position.

FIG. 4 is a side view representation of the centralising device, with the arms of the centraliser of the centralising device in a radially inwards position.

FIG. 5 is a cross-sectional view of the centralising device on line I-I shown in FIG. 3.

FIG. 6 is a part cross-sectional view on line I-I of the centralising device shown in FIG. 3, illustrating an upper end and a lower end of a pressure lubrication system of the device of FIG. 3. In FIG. 6 a resiliently deformable member of the pressure lubrication system is shown in a compressed state.

FIG. 7 is a part cross-sectional view of the centralising device on line I-I shown in FIG. 3. In FIG. 7 the resiliently deformable member of the pressure lubrication system is shown in an uncompressed state.

BEST MODES FOR CARRYING OUT THE INVENTION

FIG. 1 provides a schematic representation of a well site **100**. A logging tool string **101** is lowered down the wellbore **102** on a wireline **103**. Wellsite surface equipment includes sheave wheels **104** typically suspended from a derrick and a winch unit **105** for uncoiling and coiling the wireline to and from the wellbore, to deploy and retrieve the logging tool **101** to and from the wellbore to perform a wellbore wireline logging operation. The logging tool string **101** comprises one or more logging tools each carrying one or more sensors or sampling tools coupled together to form the logging tool string **101**. The wireline **102** includes a number of wires or cables to provide electrical power to the one or more sensors and transmit sensor data to the wellsite surface. One or more centralising or other conveyance devices **110** may be provided along the logging tool **101** to centralise and convey the logging tool **101** in the wellbore **102**.

The tool string **101** comprises a sensor **106** at the bottom end of the tool string. Sensor **106** is mounted on a rotating head **107** at the bottom end of the tool string **101**. The rotating head **107** spins and enables the sensor **106** to record azimuthal signals from the casing wall, cement sheath, and/or wellbore wall as the tool string **101** is slowly winched out of the wellbore **102**. The rotating head **107** and sensor **106** together form a rotating sensor assembly. The tool string **101** comprises a drive or motor **108** to drive rotation of the rotating sensor assembly. According to the present invention, a centralising device **1** is provided at the bottom end of the tool string **101** below the rotating sensor assembly. The drive **108** is in the tool string **101** above the rotating sensor assembly **106**, **107**, and the centralising device **1** is located below the rotating sensor assembly **106**, **107**.

FIGS. 3 to 5 provide illustrate the centralising device **1**. The centralising device **1** comprises a centraliser **2** mounted on a rotating shaft **4**. The shaft **4** is fixed to the rotating head **107**, as shown in FIG. 2, to rotate together with the rotating head **107** and sensor **106**. The shaft **4** may comprise a flange **4a** or other mount configured to fix the shaft **4** to the rotating head **107**. Alternatively, the shaft may be integral with the rotating head. The flange may be fixed to the rotating head by fasteners such as bolts (not shown). The shaft **4** has a central longitudinal axis ‘A’ colinear with a central rotational axis of the rotating head **107**. Such that the shaft **4** and rotating head **107** rotate together on the rotational axis A. An upper end of the shaft **4** is fixed to the rotating sensor assembly **107**, **106**. The shaft is cantilevered from the

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rotating sensor assembly. The upper end of the shaft 4 is connected to the rotating sensor assembly. The bottom end of the shaft 4 is the terminal end.

The centraliser 2 is rotationally mounted on the rotating shaft 4 to allow for rotation of the shaft 4. The shaft 4 can rotate without rotation of the centraliser 2. The rotational mounting of the centraliser 2 on the shaft 4 allows for the centraliser 2 to be rotationally static in the bore 102 (i.e. to not rotate relative to the bore 102). The centraliser 2 is free to rotate in the bore 102. However, the centraliser 2 is rotationally decoupled from the rotating shaft 4 and rotating head 107 and sensor 106 such that the centraliser 2 does not rotate with the shaft 4. Thus, the centraliser 2 can remain rotationally stationary in the bore 102 as the device 1 traverses the wellbore.

The centraliser 2 comprises a plurality of arms 3 spaced circumferentially apart around the longitudinal central axis A of the device 1. In the illustrated embodiment there are four arms 3, however the centraliser 2 may have three, four or more arms, for example five or six arms. The arms 3 are configured to move radially to engage the wellbore wall 102a to provide a centering force to maintain the shaft and therefore the rotating head 107 and sensor 106 at the centre of the wellbore 102. The centraliser 2 supports the shaft 4 between the upper end of the shaft and the lower end of the shaft. The upper end of the shaft is connected to the rotating head, and the lower end is the terminal end of the shaft, i.e. the terminal end is at the bottom end of the tool string.

In the illustrated embodiment each arm 3 is an arm assembly (or linkage assembly) comprising a first arm or link 5 and a second arm or link 6. The first arm 5 is pivotally connected to a first support member 7 by a first pivot joint 9, and the second arm 6 is pivotally connected to a second support member 8 by a second pivot joint 10. The first and second arms 5, 6 are pivotally attached together by a third pivot joint 11. Each pivot joint 9, 10, 11 has a pivot pin or axle on which the arms 5, 6 pivot about a pivot axis 9a, 10a, 11a, being an axis of the pin or axle.

In the illustrated embodiment each arm assembly 3 comprises a roller or wheel 12 located at the third pivot joint 11 to contact the wellbore wall 102a. In use the arm assemblies 3 are biased radially outwards so that the wheels 12 contact the wellbore wall to reduce friction between the wellbore wall 102a and the tool string 101 as the tool string 101 traverses the well bore 102. The wheel 12 may have a rotational axis colinear with the third pivot axis 11a of the third pivot joint 11, i.e. the wheel rotates around the third pivot joint. Alternatively, the wheel may be rotationally coupled to the first or second arm on a rotational axis adjacent to the third pivot axis 11a.

One or both support members 7, 8 are adapted to move axially along the axis A. Axial movement of one or both support members allows each arm assembly 3 to move radially to engage the wellbore wall 102 by pivoting of the first, second and third pivot joints 9, 10, 11. One or both support members 7, 8 may comprise a collar or annular member colinear with the shaft axis A.

The illustrated centraliser 2 is provided by way of example. Other alternative centralisers as known in the art may be provided to the rotating shaft 4. For example, in an alternative embodiment, each arm 3 is or comprises a bow spring connected between the first and second support members 7, 8.

The support members 7, 8 are mounted relative to the shaft 4 allow for rotation of the shaft 4 within each support member 7, 8. The rotational mounting of the support members 7, 8 with respect to the rotating shaft 4 allows for the

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centraliser 2 to be rotationally static in the bore 102 as described above. The support members are not rotationally keyed relative to the rotational shaft, to allow for relative rotation therebetween. Thus, the support members 7, 8 and arms 3 can remain rotationally static in the bore 102.

The centraliser 2 has one or more spring elements 13 to provide a force to the arms to force the arms 3 against the wellbore wall 102a to provide a centralising force to maintain the centralising device and therefore rotating sensor 106 centrally within the wellbore 102. In the illustrated embodiment, the centraliser 2 comprises leaf springs 13 (refer FIG. 5) acting on the second arm 6 to bias the arm assemblies 3 radially outwards against the wellbore wall 102a. Alternative spring arrangements may be provided, such as one or more axial springs acting on one or both support members 7, 8 to bias the support members 7, 8 axially inwards, thereby biasing the arms 3 radially outwards. Alternatively, as noted above, the arms may be bow-springs, where the bow-springs function as both the arms and spring elements. The springs are the only means of powering the arms radially outwards. There is no other power input to the device. The centraliser is therefore a passive device, with energisation of the arms radially outwards being provided only by the one or more springs.

In the illustrated embodiment, the leaf springs act on the second arms 6. The second arms 6 are longer than the first arms 5 to accommodate the radially acting springs 13. The centraliser 2 is arranged with the shorter first arms 5 upper most, nearest to the rotating sensor 106, to position the wheels 12 of the centraliser 2 as close to the sensor 106 as possible. Centralisation of the sensor 106 may be improved by placing the centraliser contact point with the wellbore (the wheels) as close to the sensor as possible.

The centraliser 2 comprises mechanical stops 14. The stops 14 set a maximum diameter for the centraliser 2. Each stop 14 limits inwards axial movement of the respective support member 7, 8, to limit the radial outward movement of the arms 3, whereby setting the maximum OD of the device.

As described above, the support members 7, 8 are mounted relative to the shaft to slide axially along the longitudinal axis A. In some embodiments, the support members may be mounted directly to the rotating shaft 4 to slide thereon. In the illustrated embodiment, and as best shown in FIG. 5, the centralising device 1 comprises a sleeve 15 rotationally mounted to the rotating shaft 4. The sleeve 15 is rotationally mounted on the rotating shaft 2 to allow for rotation of the shaft 4 within the sleeve 15. The shaft can rotate without rotation of the sleeve. The centraliser 2 is mounted to the sleeve 15, thereby rotationally mounting the centraliser 2 to the shaft 4. The centraliser 2, including arms 3, support members 7, 8, springs 13 and mechanical stops 14, is mounted to the sleeve 15, to rotationally mount the centraliser 2 to the shaft. This allows for the centraliser 2 to be rotationally static in the bore 102, as described above. The support members 7, 8 may be free to rotate on the sleeve 15. Alternatively, the support members 7, 8 may be keyed to the sleeve to prevent relative rotation therebetween. One or both stops 14 may be fixed to the sleeve 15 or may be integrally formed with the sleeve 15.

As best shown in FIG. 6, in the illustrated embodiment the sleeve 15 is mounted to the rotating shaft 4 via bearings 16, 17. One bearing 16 is located at or towards an upper end of the shaft 4 (the end connected to the rotating head 107) and the other bearing 17 is located at or towards the terminal lower end of the shaft. Each bearing 16, 17 may for example comprise roller elements, such as ball bearings or roller

bearings (roller elements not illustrated). The bearings **16**, **17** may comprise an inner race mounted to the shaft and an outer race mounted to the sleeve with roller elements between the races. Alternatively, the bearings **16**, **17** may comprise roller elements such as balls or pins captured between the sleeve and shaft, i.e. the sleeve and shaft provide the inner and outer races. The bearings **16**, **17** provide for a low friction rotating interface between the sleeve **15** and the shaft **4**.

The illustrated centralising device **1** comprises a pressure compensated bearing lubrication system **20**. The lubrication system **20** provides lubricant to the bearings **16**, **17** at a pressure greater than an ambient pressure in the wellbore (the wellbore pressure or ambient wellbore pressure).

With reference to FIGS. **6** and **7**, the lubrication system comprises a housing **21** within which is provided a resiliently deformable member **22**. The resiliently deformable member **22** is sealingly attached to the housing **21** at mounting flange **23**. The resiliently deformable member has a mounting flange **23** to mount the resiliently deformable member to the housing. The flange may provide a seal with the housing, or a seal may be provided between the flange and housing. The housing **21** is attached to the sleeve **15**. The housing **21** is attached to a lower end of the sleeve **15**. Such that an inside of the housing is in communication with an inside of the sleeve. The housing **21** with resiliently deformable member **22** closes an end of the sleeve **15**, to cover a lower end of the shaft **4**. An opposite upper end of the sleeve **15** is sealed by a seal **26** acting between the rotating shaft **4** and the sleeve **15**. For example, the seal **26** may comprise a lip seal or mechanical seal. With the resiliently deformable member **22** mounted to the housing **21**, the resiliently deformable member **22** seals or closes the housing **21** to provide a sealed volume **25** defined by the resiliently deformable member, an inner surface of the housing **21**, an annular space between the sleeve **21** and the rotating shaft **4**, and the seal **26**. Thus, the sealed volume **25** extends between the resiliently deformable member **22** and the seal **26**. The sealed volume **25** is filled with lubricant in use. The bearings **16** and **17** are received within the sealed volume and are therefore immersed in the lubricant received in the sealed volume **25**. In FIG. **6** the bearings are shown schematically, however, one skilled in the art will understand a bearing assembly comprising roller elements such as balls has a flow path for lubricant to flow through the bearings. The resiliently deformable member **22** provides a movable interface between the wellbore fluid and the lubricant received in the sealed volume **25** to pressure compensate the lubricant relative to the ambient pressure of the wellbore.

In the illustrated embodiment, the resiliently deformable member comprises a bellows formation **22**. The bellows formation has an open end **27** and a closed end **28**. A flexible member **30**, for example a sheet of rubber or resilient diaphragm such as an elastomeric diaphragm, is sealingly connected to or over the open end **27** of the bellows formation.

FIG. **6** shows the bellows **22** in a compressed state and the flexible member in an inflated state. FIG. **7** shows the bellows **22** in an expanded state and the flexible member in an uninflated state. The flexible member **30** may be mounted or connected to the housing **21**, to provide a second volume or a chamber **29** defined by an interior of the bellows **22**, and an inner surface of the flexible member **30**. A cover **31** is preferably provided over the flexible member **30** and clamps the flexible member **30** to the housing **21**. The cover **31** is provided with at least one opening **32** such that the external

surface of the flexible member **30** is in communication with the wellbore fluids that surround centralising device **1** in use. The cover may be fixed to the housing by fasteners such as screws (e.g. screws **32**, FIG. **4**).

A substantially incompressible fluid, for example silicone oil, is provided in the chamber **29** defined (at least in part) by the interior of the bellows formation **22** and the inner surface of the flexible member **30**. The oil is substantially incompressible, such that any expansion or contraction of the bellows formation **22** causes deflection of the flexible member **30** and vice versa. In operation, deflection of the bellows formation may be due to thermally and pressure induced volume changes of the lubricant received in the sealed volume **25** or minor loss of lubricant through the seal **26**.

The bellows formation **22** is preferably formed from a metal, for example Inconel or stainless steel. In preferred embodiments multiple annular metal rings **34** are welded together to form the bellows formation **22**. The metal rings may be beveled metal rings. The bellows formation is preferably a spring bellows formation so that a force is required to compress the bellows from an expanded configuration to a compressed or less expanded configuration. The formation is preferably elastically deformable along the central axis A.

The inside of the bellows may be filled with the substantially incompressible fluid and then the flexible diaphragm and cover assembled to the housing to contain the fluid in the chamber **29**.

The sealed volume **25** is defined by an exterior **24** of the bellows formation **22**, the inner surface of the housing **21**, an ID of the sleeve **15**, an OD of the rotating shaft **4**, and the seal **26**. The volume **25** is filled with a lubricant before use via a one-way valve (valve **35**, FIG. **4**). With the housing filled with lubricant, the exterior of the bellows is immersed in the lubricant. The exterior of the bellows faces the lubricant or is exposed to the lubricant. The lubricant is preferably pressurised sufficiently when filling the volume **25** to cause a compression of the bellows formation **22**. Compression of the bellows **22** causes movement of the fluid in chamber **29** to cause outwards deflection of the flexible member **30**, as shown in FIG. **6**.

The deflection of the flexible member **30** may be limited to around 10% or less elongation, to ensure that the flexible member **30** does not wear out or fail due to fatigue after repeated uses. Preferably the lubrication system can be used on multiple runs in the wellbore with minimal maintenance between runs, in other words reusable. The bellows may be damaged if wellbore cuttings become lodged between the metal rings that form the spring bellows. By providing the flexible member **30** to define a chamber **29** with the interior of the bellows, and filling the chamber **29** with a fluid, the bellows **22** is separated from the wellbore fluid and wellbore cuttings and debris. Thus, the lubrication system comprising the flexible member and chamber filled with fluid prevents wellbore cuttings and debris from interfering with the bellows formation.

In use, the ambient pressure of the wellbore fluid surrounding the tool string and centralising device **1** bears on the outer surface of the flexible member **30**. The flexible member **30** deflects under the ambient pressure, transferring the pressure through the fluid in chamber **29** to the interior of the bellows formation **22**. The bellows formation is free to expand axially to allow the transfer of pressure to the lubricant in the sealed volume **25**. In use, the bellows formation is elastically compressed and consequently provides additional pressure to the lubricant in the sealed

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volume **25**. In other words, the bellows provides a bias force against the lubricant in the housing. The bellows is biased to an expanded configuration. For example, the bellows is constructed in an expanded configuration, for example as in FIG. 7, and a force is required to deflect the bellows formation from the expanded configuration to a compressed or less expanded configuration, for example as in FIG. 6. In this way the pressure of the lubricant within the bearings **16**, **17** and on the inside of the shaft seal **26** is kept at a slightly higher pressure (for example around 20 psi higher) than the ambient wellbore pressure on the outside the seal **26**, regardless of any change in wellbore pressure. In this way, the bearings remain immersed in the lubricant, to ensure a low friction rotational interface between the rotating shaft **4** and the sleeve and centraliser **2**. A low friction rotational interface is important given the rotation of shaft **4** is driven by a drive mechanism **108** designed to drive rotation of the rotating head **107**. Friction between the rotating shaft **4** and centraliser **2** introduces additional load that the rotating head drive **108** must overcome.

In an alternative embodiment the resiliently deformable member may comprise the flexible member **30** (e.g. without the inclusion of a bellows formation **22**). For example, a sheet of rubber or resilient diaphragm such as an elastomeric diaphragm may provide the movable interface between the wellbore fluid and the lubricant in the sealed volume **25**. In such an embodiment, the sealed volume **25** may be defined by an inner side of the flexible member **30**, the inner surface of the housing **21**, an ID of the sleeve **15**, an OD of the rotating shaft **4**, and the seal **26**. In use, the ambient pressure of the wellbore fluid surrounding the tool string and centralising device **1** bears on the outer surface of the flexible member **30**. The flexible member **30** deflects under the ambient pressure, transferring the pressure to the lubricant in the sealed volume **25**. Before use, the sealed volume is filled with lubricant via valve **35** (FIG. 4). The lubricant is preferably pressurised sufficiently when filling the volume **25** to cause the flexible member to elastically expand. Elastic expansion of the flexible member provides a bias to pressurise the lubricant. Thus, the pressure of the lubricant within the bearings **16**, **17** and on the inside of the shaft seal **26** is kept at a slightly higher pressure (for example around 5 psi higher) than the ambient wellbore pressure on the outside the seal **26**, regardless of any change in wellbore pressure.

The compensated lubrication system is practically frictionless and responds immediately to changes in wellbore pressure to maintain a small positive pressure differential between the lubricant and the well bore environment. A small pressure differential (for example less than 20 psi) is optimal to minimise friction of the seal **26** between the rotating shaft **4** and the sleeve **15**.

In the illustrated embodiment the centraliser **2** is mounted on a single sleeve **15**. Alternatively, each support member may be mounted on a separate sleeve. In a further alternative embodiment, each support member may be mounted directly to the shaft **4**. However, such alternative embodiments introduce complexity where a bearing system is desirable between the centraliser **2** and rotating shaft **4**, and in particular a bearing system immersed in a compensated lubrication system.

Centralisation is improved by locating the centraliser below the rotating sensor. The centraliser **2** is positioned close to the sensor **106**, precisely where centralisation is needed, to ensure high quality data. Further, another centraliser **110** may be located above the rotating sensor **106**, such that the rotating sensor assembly **106**, **107** is supported

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by a centraliser (**110**, FIG. 1) positioned above and the centraliser **2** below the rotating sensor assembly. Locating the centraliser on a rotating shaft connected to the rotating head **107** below the sensor **106** at the very bottom of the tool string may improve centralisation of the sensor **106**, particularly in curved or dog-logged sections of the wellbore **102**.

The invention has been described with reference to centering a tool string in a wellbore during a wireline logging operation. However, a centralising device according to the present invention may be used for centering a rotating sensor assembly in a bore in other applications, for example to center a rotating camera in a pipe for inspection purposes.

Although this invention has been described by way of example and with reference to possible embodiments thereof, it is to be understood that modifications or improvements may be made thereto without departing from the spirit or scope of the appended claims.

The invention claimed is:

1. A device for centering a rotating sensor assembly in a wellbore, the rotating sensor assembly located at the bottom end of a tool string, the device comprising:

a centraliser comprising a plurality of arms spaced circumferentially apart around a longitudinal central axis of the device,

a shaft with a mount for fixing the shaft to the rotating sensor assembly to extend below the rotating sensor assembly, so that the shaft and rotating sensor assembly rotate together on the longitudinal central axis,

wherein the centraliser is rotationally mounted to the shaft to allow the shaft to rotate within the centraliser and with the centraliser positioned below the rotating sensor assembly, and

in use, an upper end of the shaft is connected to the rotating sensor assembly, and the lower end of the shaft is a terminal end at the bottom end of the tool string.

2. The device as claimed in claim 1, wherein the centraliser supports the shaft between the upper connected end of the shaft and the lower terminal end of the shaft.

3. The device as claimed in claim 2, wherein the centraliser comprises a first support member and a second support member, the arms connected between the first and second support members,

wherein one or both the first and second support members are adapted to move axially along the central longitudinal axis, and

wherein the first and second support members are rotationally mounted to the shaft.

4. The device as claimed in claim 1, wherein the centralising device comprises a sleeve rotationally mounted to the shaft to allow for rotation of the shaft within the sleeve, the centraliser mounted to the sleeve to thereby rotationally mount the centraliser to the shaft.

5. The device as claimed in claim 4, wherein the device comprises an upper bearing located at or towards an upper end of the shaft and a lower bearing located at or towards the lower end of the shaft, the upper and lower bearings between the shaft and sleeve.

6. The device as claimed in claim 5, wherein the device comprises a pressure compensated bearing lubrication system, the pressure compensated bearing lubrication system comprising:

a housing attached to a lower end of the sleeve,

a resiliently deformable member attached to the housing, and

a seal between the rotating shaft and the sleeve at the upper end of the sleeve,

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a sealed volume defined by the resiliently deformable member, an inside of the housing, an OD of the shaft, an ID of the sleeve, and the seal, the upper and lower bearings received in the sealed volume,

in use the sealed volume filled with lubricant, the resiliently deformable member providing a movable interface between the wellbore fluid and the lubricant received in the sealed volume to pressure compensate the lubricant relative to the ambient pressure of the wellbore.

7. The device as claimed in claim 6, wherein the housing with the resiliently deformable member closes an end of the sleeve to cover a lower end of the shaft.

8. The device as claimed in claim 6, wherein the pressure compensated bearing lubrication system comprises a cover over the resiliently deformable member, the cover with at least one opening such that well bore pressure is applied to the resiliently deformable member.

9. The device as claimed in claim 6, wherein the resiliently deformable member is or comprises a bellows formation or a flexible member.

10. The device as claimed in claim 9, wherein the resiliently deformable member is the bellows formation and the sealed volume is defined by an exterior of the bellows formation, the inside of the housing, the ID of the sleeve, the OD of the rotating shaft, and the seal.

11. The device as claimed in claim 9, wherein the pressure compensated bearing lubrication system comprises a flexible member,

The bellows formation has an open end and a closed end, the flexible member sealingly connected to or over the open end of the bellows formation to form a chamber defined at least in part by an interior of the bellows formation and an inner surface of the flexible member, and

In use the chamber filled with a fluid so that wellbore fluid acting on an external surface of the flexible member is communicated to the bellows formation.

12. The device as claimed in claim 1, wherein the centraliser comprises a first support member and a second support member axially spaced apart along a longitudinal axis of the device; and

each arm is an arm assembly, the plurality of arm assemblies connected between the first and second support members, each arm assembly comprising:

a first arm pivotally connected to the first support member by a first pivot joint,

a second arm pivotally connected to the second support member by a second pivot joint, the first and second arms pivotally connected via a third pivot joint,

wherein one or both of the first and second support members is adapted to move axially along the longitudinal axis to allow the arm assemblies to extend and retract radially with respect to the longitudinal axis.

13. The device as claimed in claim 12, wherein each arm assembly comprises a roller or wheel to contact the wellbore wall.

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14. The device as claimed in claim 12, wherein the centraliser comprises one or more spring elements to bias the arm assemblies radially outwards to contact the bore wall.

15. The device as claimed in claim 1, wherein the centraliser is a passive device, with energisation of the arms radially outwards being provided by one or more spring elements of the device only.

16. A device for centering a rotating sensor assembly in a wellbore, the rotating sensor assembly located at the bottom end of a tool string, the device comprising:

a centraliser comprising a plurality of arms spaced circumferentially apart around a longitudinal central axis of the device,

a shaft with a mount for fixing the shaft to the rotating sensor assembly to extend below the rotating sensor assembly, so that the shaft and rotating sensor assembly rotate together on the longitudinal central axis,

wherein the centraliser is rotationally mounted to the shaft to allow the shaft to rotate within the centraliser and with the centraliser positioned below the rotating sensor assembly, and

a sleeve rotationally mounted to the shaft to allow for rotation of the shaft within the sleeve, the centraliser mounted to the sleeve to thereby rotationally mount the centraliser to the shaft, and

an upper bearing located at or towards an upper end of the shaft and a lower bearing located at or towards the lower end of the shaft, the upper and lower bearings between the shaft and sleeve.

17. The device as claimed in claim 16, wherein the device comprises a pressure compensated bearing lubrication system, the pressure compensated bearing lubrication system comprising:

a housing attached to a lower end of the sleeve, a movable interface mounted to the housing, and a seal between the rotating shaft and the sleeve at the upper end of the sleeve,

a sealed volume defined by the moveable interface, an inside of the housing, an OD of the shaft, an ID of the sleeve, and the seal, the upper and lower bearings received in the sealed volume,

in use the sealed volume filled with lubricant, wherein wellbore fluid acting on the movable interface pressure compensates the lubricant received in the sealed volume relative to the ambient pressure in the wellbore.

18. The device as claimed in claim 16, wherein the housing with the movable interface closes an end of the sleeve to cover a lower end of the shaft.

19. The device as claimed in claim 18, wherein the pressure compensated bearing lubrication system comprises a cover over the moveable interface, the cover with at least one opening such that well bore pressure is applied to the moveable interface.

20. The device as claimed in claim 18, wherein the movable interface is or comprises a bellows formation or a flexible member.

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